NAME:-Sheikh Wasimuddin

ROLL NO:-73

BRANCH :-CSE(IOT)

LAB 06	
Aim:	Implementation of Data Structure in R.
Theory	A data structure is a particular way of organizing data in a computer so that it can be used effectively. The idea is to reduce the space and time complexities of different tasks. Data structures in R programming are tools for holding multiple values. R's base data structures are often organized by their dimensionality (1D, 2D, or nD) and whether they're homogeneous (all elements must be of the identical type) or heterogeneous (the elements are often of various types). This gives rise to the six data types which are most frequently utilized in data analysis. The most essential data structures used in R include: • Vectors • Lists • Dataframes • Matrices • Arrays • Factors • Tibbles • **1. Vectors** A vector is an ordered collection of basic data types of a given length. The only key thing here is all the elements of a vector must be of the identical data type e.g homogeneous data structures. Vectors are one-dimensional data structures. **2. Lists** A list is a generic object consisting of an ordered collection of objects. Lists are heterogeneous data structures. These are also one-dimensional data structures. A list can be a list of vectors, list of matrices, a list of characters and a list of functions and so on. **3. Dataframes** Dataframes are generic data objects of R which are used to store the tabular data. Dataframes are the foremost popular data objects in R programming because we are comfortable in seeing the data within the tabular form. They are two-dimensional, heterogeneous data structures. These are lists of vectors of equal lengths. Data frames have the following constraints placed upon them: • A data-frame must have column names and every row should have a unique name. • Each column must have the identical number of items. • Each item in a single column must be of the same data type. • Different columns may have different data types.

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	To create a data frame we use the data.frame() function. **4. Matrices** A matrix is a rectangular arrangement of numbers in rows and columns. In a matrix, as we know rows are the ones that run horizontally and columns are the ones that run vertically. Matrices are two-dimensional, homogeneous data structures. Now, let's see how to create a matrix in R. To create a matrix in R you need to use the function called matrix. The arguments to this matrix() are the set of elements in the vector. You have to pass how many numbers of rows and how many numbers of columns you want to have in your matrix and this is the important point you have to remember that by default, matrices are in column-wise order. **5. Arrays** Arrays are the R data objects which store the data in more than two dimensions. Arrays are n-dimensional data structures. For example, if we create an array of dimensions (2, 3, 3) then it creates 3 rectangular matrices each with 2 rows and 3 columns. They are homogeneous data structures. Now, let's see how to create arrays in R. To create an array in R you need to use the function called array(). The arguments to this array() are the set of elements in vectors and you have to pass a vector containing the dimensions of the array. **6. Factors** Factors are the data objects which are used to categorize the data and store it as levels. They are useful for storing categorical data. They can store both strings and integers. They are useful to categorize unique values in columns like "TRUE" or "FALSE", or "MALE" or "FEMALE", etc.. They are useful in data analysis for statistical modeling. Now, let's see how to create factors in R. To create a factor in R you need to use the function called factor(). The argument to this factor() is the vector. **7. Tibbles** Tibbles are an enhanced version of data frames in R, part of the tidyverse. They offer improved printing, stricter column types, consistent subsetting behavior, and allow variables to be referred to as objects. Tibbles provide a modern, user-friendly approach to tabular data in R. Now, let's see how we can create a tibble in R. To create tibbles in R we can use the tibble function from the tibble package, which is part of the tidyverse.
Program:	1) #Sheikh Wasimuddin 23071044 IOT-73 # R program to illustrate Vector

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```
X = c(1, 3, 5, 7, 8)

print(X)
```

2) #Sheikh Wasimuddin 23071044 IOT-73

```
# R program to illustrate a List

empId = c(1, 2, 3, 4)

empName = c("Tanmay", "Sheikh", "Vedant", "Tejas")

numberOfEmp = 4

empList = list(empId, empName, numberOfEmp)

print(empList)
```

3) #Sheikh Wasimuddin 23071044 IOT-73

```
# R program to illustrate dataframe

Name = c("Vedant", "Tanmay", "Sheikh")

Language = c("R", "Python", "Java")

Age = c(20, 19, 19)

df = data.frame(Name, Language, Age)

print(df)
```

4) #Sheikh Wasimuddin 23071044 IOT-73

```
# R program to illustrate a matrix

A = matrix(

  c(10, 20, 30, 40, 50, 60, 70, 80, 90),

  nrow = 3, ncol = 3,

  byrow = TRUE

)
```

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```
  print(A)
```

5) #Sheikh Wasimuddin 23071044 IOT-73

```
  # R program to illustrate an array
  A = array(
    c(10, 29, 38, 47, 55,96, 87, 58),
    dim = c(2,2, 2)
  )
  print(A)
```

6) #Sheikh Wasimuddin 23071044 IOT-73

```
  # R program to illustrate factors
  # Creating factor using factor()
  fac = factor(c("Male", "Female", "Male",
               "Male", "Female", "Male", "Female"))
  print(fac)
```

7) #Sheikh Wasimuddin 23071044 IOT-73

```
# Load the tibble package
my_data <- tibble(
  name = c("sheikh","Tanmay", "Vedant", "Tejas"),
  age = c(20,20, 21, 20),
  city = c("Bilaspur","Nagpur", "Pauni", "Nanded")
)
```

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	print(my_data)
Output:	1) Untitled1* Source on Save 1 #Sheikh Wasimuddin 23071044 IOT-73 2 X = c(1, 3, 5, 7, 8) 3 print(X) 4 4:1 (Top Level) Console Terminal Background Jobs R 4.4.2 · ~/ You are welcome to redistribute it under certain conditions. Type 'license()' or 'licence()' for distribution details. Natural language support but running in an English locale R is a collaborative project with many contributors. Type 'contributors()' for more information and 'citation()' on how to cite R or R packages in publications. Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R. > source("~/.active-rstudio-document") [1] 1 3 5 7 8 > 2)

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Untitled1*

Source on Save

```r
#Sheikh Wasimuddin 23071044 IOT-73
# R program to illustrate a List
empId = c(1, 2, 3, 4)
empName = c("Tanmay", "Sheikh", "Vedant", "Tejas")
numberOfEmp = 4
empList = list(empId, empName, numberOfEmp)
print(empList)
```

1:35 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
help.start()  for an HTML browser interface to help.
Type 'q()' to quit R.

> source("~/.active-rstudio-document")
[1] 1 3 5 7 8
> source("~/.active-rstudio-document")
[[1]]
[1] 1 2 3 4

[[2]]
[1] "Tanmay" "Sheikh" "Vedant" "Tejas"

[[3]]
[1] 4
```

3)

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Untitled1*

Source on Save

```r
#Sheikh Wasimuddin 23071044 IOT-73
# R program to illustrate dataframe
Name = c("Vedant", "Tanmay", "Sheikh")
Language = c("R", "Python", "Java")
Age = c(20, 19, 19)
df = data.frame(Name, Language, Age)
print(df)
```

8:1 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
[[1]]
[1] 1 2 3 4

[[2]]
[1] "Tanmay" "Sheikh" "Vedant" "Tejas"

[[3]]
[1] 4

>
> source("~/.active-rstudio-document")
     Name Language Age
1 Vedant        R  20
2 Tanmay   Python  19
3 Sheikh     Java  19
>
```

4)

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Untitled1*

Source on Save

```r
#Sheikh Wasimuddin 23071044 IOT-73
# R program to illustrate a matrix
A = matrix(
  c(10, 20, 30, 40, 50, 60, 70, 80, 90),
  nrow = 3, ncol = 3,
  byrow = TRUE
)
print(A)
```

9:1 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
[[3]]
[1] 4

>
> source("~/.active-rstudio-document")
    Name Language Age
1 Vedant        R  20
2 Tanmay   Python  19
3 Sheikh     Java  19
> source("~/.active-rstudio-document")
     [,1] [,2] [,3]
[1,]   10   20   30
[2,]   40   50   60
[3,]   70   80   90
>
```

5)

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Untitled1*

Source on Save

```
#Sheikh Wasimuddin 23071044 IOT-73
# R program to illustrate an array
A = array(
  c(10, 29, 38, 47, 55,96, 87, 58),
  dim = c(2,2, 2)
)
print(A)
```

8:1 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
[2,]    40    50    60
[3,]    70    80    90
> source("~/.active-rstudio-document")
, , 1

     [,1] [,2]
[1,]   10   38
[2,]   29   47

, , 2

     [,1] [,2]
[1,]   55   87
[2,]   96   58
```

6)

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Untitled1*

Source on Save

```r
#Sheikh Wasimuddin 23071044 IOT-73
# R program to illustrate factors
# Creating factor using factor()
fac = factor(c("Male", "Female", "Male",
               "Male", "Female", "Male", "Female"))
print(fac)
```

7:1 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
, , 1

     [,1] [,2]
[1,]   10   38
[2,]   29   47

, , 2

     [,1] [,2]
[1,]   55   87
[2,]   96   58

> source("~/.active-rstudio-document")
[1] Male   Female Male   Male   Female Male   Female
Levels: Female Male
>
```

7)

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Untitled1*

```r
#Sheikh Wasimuddin 23071044 IOT-73
# Load the tibble package
my_data <- tibble(
  name = c("sheikh","Tanmay", "Vedant", "Tejas"),
  age = c(20,20, 21, 20),
  city = c("Bilaspur","Nagpur", "Pauni", "Nanded")
)
print(my_data)
```

2:26 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
A tibble: 4 x 3
  name      age city
  <chr>   <dbl> <chr>
1 sheikh     20 Bilaspur
2 Tanmay     20 Nagpur
3 Vedant     21 Pauni
4 Tejas      20 Nanded
> source("~/.active-rstudio-document")
A tibble: 4 x 3
  name      age city
  <chr>   <dbl> <chr>
1 sheikh     20 Bilaspur
2 Tanmay     20 Nagpur
3 Vedant     21 Pauni
4 Tejas      20 Nanded
>
```

Conclusion	Implementation of Data Structure in R executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar